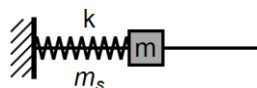


Chap. 21 The Harmonic Oscillator

1. Massive Spring (I): Effective Mass

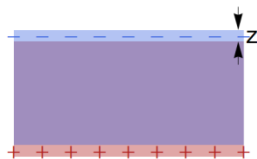
If the mass of a spring m_s is not negligible but is small compared to the mass m of the object attached to it. Let k be the (static) force constant, what is the period T of motion for the mass m ?



Note: Take into account the mass of the spring, the oscillation period is longer than $T_0 = 2\pi\sqrt{m/k}$. We can introduce the effective mass m^* from $T = 2\pi\sqrt{m^*/k}$. We can see the massive spring effectively increase the mass of the oscillator.

2. Plasma Oscillation (I): Plasma Frequency

Plasma oscillation means a kind of collective oscillation of charged particles in the material, typically metals. When an external field is switched on, all electrons in the metal will displace a distance z relative to the positive ions. There will be the positive and negative charge accumulations on the opposite sides of the metal, as shown in the following figure.



(a) Given the electron number density n in the metal, what is the surface charge density (charge per surface area) on both sides?

The electric field induced by the charged layers balances the external field such that the system is in equilibrium. When the external field is switched off, the balance is broken, and the electrons are to move under the influence of the electric field of the charged layers.

(b) How much electric force will be exerted on the electron inside the metal?

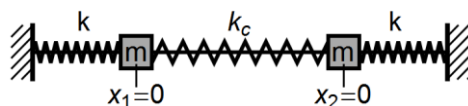
It turns out that the electric force is a restoring force that leads to the collective oscillation of electrons around their equilibrium positions.

(c) Given the effective mass m^* and electric charge e of the electron, what is its oscillating (circular) frequency ω_p ?

Note: ω_p is known as the plasma frequency.

3. Coupled Oscillator

We have two spring and mass oscillators with stiffness constants k , and masses m coupled together by a third spring of stiffness k_c . It should be noted that $k_c \ll k$, so we can say that our system is weakly coupled. Let us ignore any damping forces, such as friction or air-resistance. And the motion of the masses is restricted to one dimension. We will make the same assumption that the springs obey Hooke's law and thus exert a linear restoring force on the masses.



Now we would like to find the general solution for the motion of the two masses. Their positions are given by $x_1(t)$ and $x_2(t)$.

(a) Write down the equations of motion for the two masses by applying Newton's second law.

There are two normal modes of vibration for this system. The symmetric normal mode occurs when both masses are displaced an equal amount in the same direction and released from rest. The anti-symmetric mode occurs when both masses are displaced an equal distance from their equilibrium positions but in opposite directions. These two normal modes oscillates with different frequency and the general behavior of the coupled oscillator is a superposition of the two modes.

(b) Decouple the equation of motions by transforming to normal coordinates q_1 and q_2 :

$$\begin{cases} q_1 = x_1 + x_2 \\ q_2 = x_1 - x_2 \end{cases}$$

q_1 gives the displacement of the symmetric mode, and q_2 is the displacement of the anti-symmetric mode. What are the equations of motion for q_1 and q_2 ? What are the frequencies of the two normal modes respectively?

(c) Let us assume that we displace the left mass by a distance $x_1(0) = A$, and the right mass by a distance $x_2(0) = B$, and release the masses from rest $x_1'(0) = 0$, $x_2'(0) = 0$. Give the solution of $x_1(t)$ and $x_2(t)$ by superpose the normal modes.

(d) With $A = 1$ and $B = 0$, plot $x_1(t)$ and $x_2(t)$ as functions of t .

We can now clearly see that both the masses display the beating phenomenon. Notice that the peaks of x_1 correspond to the troughs of x_2 . The energy of the system is continuously transferred between the two masses, while the total energy of the system remains constant.

Chap. 23 Resonance

1. Plasma Oscillation (II): Drude Conductivity

In the uniform electric field, a free electron will be accelerated constantly. However it is not the case for electrons in the metal, due to the presence of the lattice. The scattering of the electron with the lattice leads to two effects. First the mass of the electron will be renormalized to the effective mass m^* by elastic scattering. Secondly the motion of electron will be damped due to the inelastic scattering with the lattice vibrations (the electron-phonon scattering). To describe the effect of damping, we introduced the relaxation time τ , such that without a driving force the velocity of the electron will diminish within the time scale τ following the form of exponentially decay $e^{-t/\tau}$. This model is known as the *Drude model*.

(a) What is the form of the damping force so as to produce the exponentially decay $e^{-t/\tau}$ behavior of the velocity?

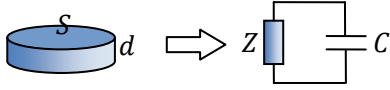
Driven by a static uniform electric field in the metal, the electron will drift with constant speed, so that the driving force will be balanced by the damping.

(b) Given the electron density n , what is the DC conductivity σ_0 of the metal according to the Drude model? Note: the conductivity is defined by $j = \sigma_0 E$, where j is the current density and E the electric field.

If the metal is in a AC circuit, the electric field in the metal will be oscillating with some frequency ω .

(c) What is the frequency-dependent (or AC) conductivity $\sigma(\omega)$? Hint: use the complex number method you have learnt in the Lecture.

Consider such metal material is made into a thin cylinder with basal area S and height d . The metal cylinder is effectively a parallel circuit of a complex impedance Z and a capacitor C , because the body of the cylinder responses to the electric field like a impedance of the conductivity $\sigma(\omega)$, while the surface of the cylinder stores electric charges like a capacitor.



(d) Show that the capacitor of a thin cylinder is given by

$$C = \frac{\epsilon_0 S}{d}.$$

Note: You may think of why the dielectric constant we used here is the one of vacuum ϵ_0 but not the one of the metal. Because the capacitor C here only models the capacitance due to the effect of two surfaces. The capacitance due to the charge transfer in the metal lies in the impedance.

(e) Show that the Z - C circuit will oscillate with the plasma frequency ω_p in the long relaxation time limit ($\omega_p \tau \gg 1$). Note: If you have forgotten what is the plasma frequency ω_p , refer to the Problem 2 of Chap. 21.

(f) Show that if the relaxation time τ is shorter than $(1/2)\omega_p^{-1}$ the plasma oscillation will be quenched by the large damping.

2. Nuclear Magnetic Resonance

Atomic nuclei have magnetic moment μ proportional to their spin angular momentum S ,

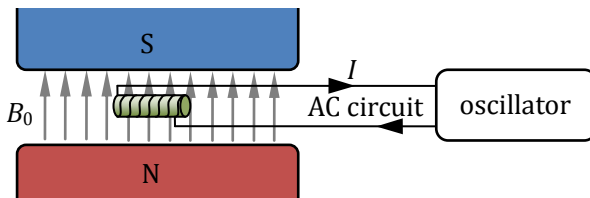
$$\mu = \gamma S.$$

The gyromagnetic ratio γ can be either positive or negative depending on the species of nucleus. As a model study, we can simply assume $\gamma > 0$ here. In the external magnetic field B , the nuclear moment encounters torque $\tau = \mu \times B$, and will precess.

(a) What is the angular velocity Ω of precession for a nucleus in the magnetic field B ?

For example a proton in a 0.5T magnetic field will precess with 21MHz frequency, about the upper limit of radio short-wave II. If the electromagnetic wave of the same frequency shines on the precessing nucleus, it will resonate with the nucleus and hence being absorbed. This phenomena is known as the nuclear magnetic resonance. F. Bloch and E. M. Purcell won 1952 Nobel Prize for the development of new method measuring nuclear magnetic resonance. Today nuclear magnetic resonance is extensively used in physics, chemistry, biophysics, medicine etc. Samples are put in perpendicular uniform magnetic field B_0 , with a weak alternating field $B_c \cos \omega_c t$ ($B_c \ll B_0$) is added horizontally. The total magnetic field can be written as

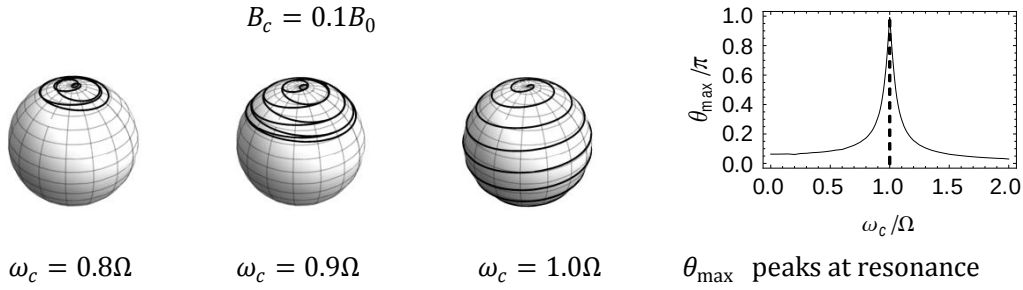
$$B = (B_c \cos \omega_c t, 0, B_0).$$



(b) Write down the equation of motion for the nuclear moment $\mu = (\mu_x, \mu_y, \mu_z)$.

(c) Design a computer program to solve the differential equation, under the condition that $B_c = 0.1B_0$, for $\omega_c = 0.8\Omega, 0.9\Omega, 1.0\Omega, 1.1\Omega$ and 1.2Ω . The initial condition is given by $\mu(t=0) = (0,0,1)$, i.e. the moment points to the north pole initially. Plot your solutions on a sphere. Measure the maximum angle $\theta_{\max}$ that the moment can tilt away from the north

pole for each ω_c . Hint: You are supposed to obtain the following figures for the first three cases.



From the computer simulation, we can see that the resonance happens when the alternating frequency ω_c of the horizontal field exactly matches the precession frequency Ω . Let us understand the physics. Since the horizontal field is oscillating, we may expect its torque on the nucleus will cancel on average. However if the circular frequency ω_c is equal to the precession frequency, the moment $\boldsymbol{\mu}$ will synchronize with the horizontal field, and gain energy from the field. The energy of the moment in the magnetic field is given by $E = -\boldsymbol{\mu} \cdot \mathbf{B}$, so as the moment becomes more and more energetic, it will flip to the opposite direction of the external field B_0 . In this case, energy absorption of $2\mu B_0$ can be observed in the AC circuit.

(d) On resonance, what is the average power of the work done by the horizontal field, when the angle between the moment $\boldsymbol{\mu}$ and the magnetic field $\mathbf{B}_0 = B_0 \mathbf{e}_z$ is θ ?

(e) How much time it takes to flip the moment from the north pole to the south pole on resonance?

In the small B_c limit ($B_c/B_0 \ll 1$), the equation of motion can be solved by perturbation approach, i.e. to solve the equation with $B_c = 0$ first then to correct the result up to the first order of B_c/B_0 .

(f) Use the perturbation approach to show that the angle θ between $\boldsymbol{\mu}$ and $\mathbf{B}_0$ follows a almost linear behavior on resonance

$$\theta = \frac{\gamma B_c}{2} \left(t + \frac{\sin 2\Omega t}{2\Omega} \right).$$